

Problem Set 73

Indeterminate Forms and L'Hopital's Rule

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Problem 2

Find the limit:

(a) $\lim_{x \rightarrow 0^+} x^{x^2}$

$$\begin{aligned} & \lim_{x \rightarrow 0^+} x^{x^2} \quad [0^0] \\ & \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} x^{x^2} \\ & \lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x^2 \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^2}} \\ & \begin{cases} \lim_{x \rightarrow 0^+} \ln x \rightarrow -\infty \\ \lim_{x \rightarrow 0^+} \frac{1}{x^2} \rightarrow \infty \end{cases} \\ & \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^2}} \stackrel{[-\infty]}{\stackrel{\infty}{\text{L'H}}} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{2}{x^3}} = \lim_{x \rightarrow 0^+} -\frac{x^2}{2} = 0 \\ & \lim_{x \rightarrow 0^+} x^{x^2} = \lim_{x \rightarrow 0^+} y = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = \boxed{1} \end{aligned}$$

(b) $\lim_{x \rightarrow 0} (1 - 2x)^{\frac{1}{x}}$

$$\begin{aligned} & \lim_{x \rightarrow 0} (1 - 2x)^{\frac{1}{x}} \quad [1^\infty] \\ & \lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} (1 - 2x)^{\frac{1}{x}} \\ & \lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 - 2x)}{x} \\ & \begin{cases} \lim_{x \rightarrow 0} \ln(1 - 2x) = 0 \\ \lim_{x \rightarrow 0} x = 0 \end{cases} \\ & \lim_{x \rightarrow 0} \frac{\ln(1 - 2x)}{x} \stackrel{[0]}{\stackrel{0}{\text{L'H}}} \lim_{x \rightarrow 0} \frac{-2}{1 - 2x} = -2 \\ & \lim_{x \rightarrow 0} (1 - 2x)^{\frac{1}{x}} = \lim_{x \rightarrow 0} y = e^{\lim_{x \rightarrow 0} \ln y} = \boxed{\frac{1}{e^2}} \end{aligned}$$

(c) $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} + \frac{5}{x^2}\right)^x$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} + \frac{5}{x^2}\right)^x \quad [1^\infty]$$

$$\begin{aligned}
\lim_{x \rightarrow \infty} y &= \lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} + \frac{5}{x^2}\right)^x \\
\lim_{x \rightarrow \infty} \ln y &= \lim_{x \rightarrow \infty} x \ln\left(1 + \frac{3}{x} + \frac{5}{x^2}\right) = \lim_{x \rightarrow \infty} \frac{\ln\left(1 + \frac{3}{x} + \frac{5}{x^2}\right)}{\frac{1}{x}} \\
&\quad \begin{cases} \lim_{x \rightarrow \infty} \ln\left(1 + \frac{3}{x} + \frac{5}{x^2}\right) = 0 \\ \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \end{cases} \\
\lim_{x \rightarrow \infty} \frac{\ln\left(1 + \frac{3}{x} + \frac{5}{x^2}\right)}{\frac{1}{x}} &\stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} \frac{\frac{-\frac{3}{x^2} - \frac{10}{x^3}}{1 + \frac{3}{x} + \frac{5}{x^2}}}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{3 - \frac{10}{x}}{1 + \frac{3}{x} + \frac{5}{x^2}} = 3 \\
\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} + \frac{5}{x^2}\right)^x &= \lim_{x \rightarrow \infty} y = e^{\lim_{x \rightarrow \infty} \ln y} = \boxed{e^3}
\end{aligned}$$

(d) $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$

$$\begin{aligned}
&\lim_{x \rightarrow \infty} x^{\frac{1}{x}} \quad [\infty^0] \\
&\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} x^{\frac{1}{x}} \\
&\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln x}{x} \\
&\quad \begin{cases} \lim_{x \rightarrow \infty} \ln x \rightarrow \infty \\ \lim_{x \rightarrow \infty} x \rightarrow \infty \end{cases} \\
&\lim_{x \rightarrow \infty} \frac{\ln x}{x} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \\
&\lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \lim_{x \rightarrow \infty} y = e^{\lim_{x \rightarrow \infty} \ln y} = e^0 = \boxed{1}
\end{aligned}$$

(f) $\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}}$

$$\begin{aligned}
&\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} \quad [1^\infty] \\
&\lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} \\
&\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln \cos x}{x^2} \\
&\quad \begin{cases} \lim_{x \rightarrow 0^+} \ln \cos x = 0 \\ \lim_{x \rightarrow 0^+} x^2 = 0 \end{cases} \\
&\lim_{x \rightarrow 0^+} \frac{\ln \cos x}{x^2} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{-\tan x}{2x} \\
&\quad \begin{cases} \lim_{x \rightarrow 0^+} -\tan x = 0 \\ \lim_{x \rightarrow 0^+} 2x = 0 \end{cases}
\end{aligned}$$

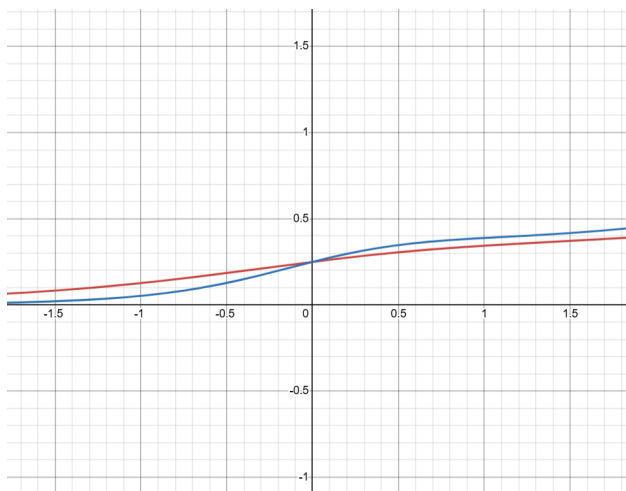
$$\lim_{x \rightarrow 0^+} \frac{-\tan x}{2x} \stackrel{\left[\frac{0}{0}\right]}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0^+} \frac{-\sec^2 x}{2} = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} y = e^{\lim_{x \rightarrow 0^+} \ln y} = \boxed{\frac{1}{\sqrt{e}}}$$

Problem 3

Let $f(x) = e^x - 1$, $g(x) = x^3 + 4x$.

(a) Illustrate L'Hopital's Rule by graphing both $\frac{f(x)}{g(x)}$ and $\frac{f'(x)}{g'(x)}$ near $x = 0$ to see that these ratios have the same limit as $x \rightarrow 0$.



(b) Calculate the exact value of the limit.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x^3 + 4x} \quad \left[\frac{0}{0}\right]$$

$$\begin{cases} \lim_{x \rightarrow 0} e^x - 1 = 0 \\ \lim_{x \rightarrow 0} x^3 + 4x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x^3 + 4x} \stackrel{\left[\frac{0}{0}\right]}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{e^x}{3x^2 + 4} = \boxed{\frac{1}{4}}$$

Problem 7

If an initial amount of money A_0 is invested at an interest rate r compounded n times a year, the value of the investment after t years is

$$A = A_0 \left(1 + \frac{r}{n}\right)^{nt}.$$

If we let $n \rightarrow \infty$, we refer to the continuous compounding of interest. Use L'Hôpital's Rule to show that if interest is compounded continuously, then the amount after t years is

$$A = A_0 e^{rt}.$$

$$\begin{aligned}
\lim_{n \rightarrow \infty} y &= \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^{nt} \quad [1^\infty] \\
\lim_{n \rightarrow \infty} \ln y &= \lim_{n \rightarrow \infty} nt \ln\left(1 + \frac{r}{n}\right) = \lim_{n \rightarrow \infty} \frac{\ln\left(1 + \frac{r}{n}\right)}{\frac{1}{nt}} \\
&\begin{cases} \lim_{n \rightarrow \infty} \ln\left(1 + \frac{r}{n}\right) = 0 \\ \lim_{n \rightarrow \infty} \frac{1}{nt} = 0 \end{cases} \\
\lim_{n \rightarrow \infty} \frac{\ln\left(1 + \frac{r}{n}\right)}{\frac{1}{nt}} &\stackrel{\substack{[\frac{0}{0}] \\ \text{L'H}}}{=} \lim_{n \rightarrow \infty} \frac{\frac{-\frac{r}{n^2}}{1 + \frac{r}{n}}}{-\frac{1}{n^2 t}} = \lim_{n \rightarrow \infty} \frac{rt}{1 + \frac{r}{n}} = rt \\
\lim_{n \rightarrow \infty} y &= e^{\lim_{n \rightarrow \infty} \ln y} = e^{rt} \\
A &= A_0 \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^{nt} = \boxed{A_0 e^{rt}}
\end{aligned}$$

Problem 8

If f' is continuous, $f(2) = 0$, and $f'(2) = 7$, evaluate

$$\lim_{x \rightarrow 0} \frac{f(2 + 3x) + f(2 + 5x)}{x}.$$

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{f(2 + 3x) + f(2 + 5x)}{x} &= \lim_{x \rightarrow 0} \frac{f(2 + 3x)}{x} + \lim_{x \rightarrow 0} \frac{f(2 + 5x)}{x} \\
&= 3 \lim_{x \rightarrow 0} \frac{f(2 + 3x) - f(2)}{3x} + 5 \lim_{x \rightarrow 0} \frac{f(2 + 5x) - f(2)}{5x} \\
&= 3f'(2) + 5f'(2) \\
&= \boxed{56}
\end{aligned}$$

Problem 9

Prove that

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^p} = 0$$

for any number $p > 0$. This shows that the logarithmic function approaches ∞ more slowly than any power of x

$$\begin{aligned}
&\begin{cases} \lim_{x \rightarrow \infty} \ln x \rightarrow \infty \\ \lim_{x \rightarrow \infty} x^p \rightarrow \infty \end{cases} \\
\lim_{x \rightarrow \infty} \frac{\ln x}{x^p} &\stackrel{[\frac{\infty}{\infty}]}{=} \lim_{x \rightarrow \infty} \frac{x^{-1}}{px^{p-1}} = \lim_{x \rightarrow \infty} \frac{1}{px^p} = \boxed{0}
\end{aligned}$$

Problem 10

Prove that

$$\lim_{x \rightarrow 0^+} x^\alpha \ln x = 0$$

for any $\alpha > 0$.

$$\begin{aligned} \lim_{x \rightarrow 0^+} x^\alpha \ln x &= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^\alpha}} \\ &\begin{cases} \lim_{x \rightarrow 0^+} \ln x \rightarrow -\infty \\ \lim_{x \rightarrow 0^+} \frac{1}{x^\alpha} \rightarrow \infty \end{cases} \\ \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^\alpha}} &\stackrel{\substack{[-\infty] \\ \infty}}{=} \lim_{x \rightarrow 0^+} \frac{x^{-1}}{-\alpha x^{-\alpha-1}} = \lim_{x \rightarrow 0^+} -\frac{x^\alpha}{\alpha} = \boxed{0} \end{aligned}$$